Can Close Social Relationship Classification Models Generalize Across Communication Datasets?

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Abstract. *Predicting close social relationships from mobile communication data can support applications such as recommendation systems, marketing, and investigative analysis. However, the scarcity of public communication datasets limits the development of robust classification models. In this context, transfer learning emerges as a potential alternative to leverage information from different communication environments. In this work, we investigate whether communication datasets generalize across domains and whether combining heterogeneous datasets improves predictive performance. Experiments were conducted using two public datasets: NetHealth and Copenhagen Networks Study. Results show that both cross-domain models and models trained on combined datasets experience substantial performance degradation when evaluated on different communication environments, revealing strong domain dependency in communication behavior.*

1. Introduction

Understanding whether two individuals share a close social relationship, such as friendship, family ties, or romantic involvement, based solely on their communication behavior is a meaningful task, as call and text interaction patterns vary systematically across relationship types and can therefore be used to infer interpersonal ties [Mäder et al. 2024]. Communication behaviors derived from phone calls and SMS exchanges reveal important indicators of social proximity, interaction intensity, and behavioral regularity, providing valuable signals for the classification of close interpersonal relationships [Chu et al. 2025].

However, communication data is difficult to obtain since, most of the times, such data is only available through controlled experiments in which participants explicitly consent to data collection and continuously log their interactions, or through highly restricted sources involving communication privacy access. As a consequence, there are very few public datasets containing detailed interpersonal communication records making it difficult to train robust machine learning models capable of generalizing across different social environments and populations.

One possible alternative to work with this limitation is the use of transfer learning strategies [Chu et al. 2025]. Instead of training and evaluating models exclusively within a single dataset, models can be trained on one communication dataset and later applied to another dataset collected in different contexts. In this work, we apply this to two public communication datasets: the NetHealth dataset from University of Notre Dame and the Copenhagen Networks Study dataset collected by University of Copenhagen.

Based on this motivation, this work examines two main research questions (RQ):

RQ1 - Cross-domain generalization. Does a communication dataset generalize to another communication dataset? This question evaluates whether a model trained on one dataset can maintain predictive performance when tested on a different dataset.

RQ2 - Effect of combining datasets from different communication environments. Can one communication dataset enrich another? This question investigates whether combining datasets during training produces better predictive performance than training using only a single dataset

2. Related Works

Recent studies have explored smartphone sensing, wearable devices, and social network analysis to infer behavioral and social patterns from human interaction data. For instance, [Tolosa et al. 2026] demonstrated that mobile sensing signals can support the prediction of mental health and behavioral indicators, while [Châtel et al. 2025] showed that social network structures can reveal relevant social and emotional patterns. Similarly, [Wang et al. 2021] investigated behavioral and activity patterns among college students using communication and sensor data.

In parallel, recent research has investigated strategies to improve predictive tasks through data enrichment and heterogeneous data integration. [Barchilon et al. 2024] explored how combining complementary datasets can improve classification performance, while [Chu et al. 2025] studied transferable social relationship inference across different urban environments, highlighting the importance of evaluating whether prediction models generalize across domains. Aligned with this perspective, our work investigates whether communication datasets generalize across domains and whether combining heterogeneous datasets improves social relationship prediction performance.

3. Data

This study was conducted using two public mobile communication datasets: the NetHealth (NH) dataset[1] from the University of Notre Dame [Wang et al. 2021] and the Copenhagen Networks Study (CNS) dataset[2] collected by the University of Copenhagen [Sapiezynski et al. 2019]. Both datasets provide mobile communication records together with self-reported social relationship information, enabling the study of close social relationship classification across different populations and environments.

For the communication data, only interactions of type phone call and SMS were considered. From these records, statistical features describing communication frequency and temporal behavior were extracted for each pair of users. For phone calls, the extracted features included the total number of calls, average call duration, minimum and maximum call duration, average call time during the day, earliest and latest call time, and the number of distinct days with communication activity. Similarly, for SMS interactions, temporal and frequency-based features were extracted, including the total number of exchanged messages, average communication time, minimum and maximum communication time, and the number of distinct communication days.

[1]NetHealth dataset: `https://sites.nd.edu/nethealth/data-2/`

[2]Copenhagen Networks Study dataset:`https://doi.org/10.6084/m9.figshare.7267433`

All communication features were computed considering both global and directional interactions (incoming and outgoing communication separately). In addition, features were also extracted independently for weekdays and weekends in order to capture differences in communication behavior across temporal contexts.

The labeled data used for close social relationship classification was obtained from self-reported relationship information available in each dataset. In the NH dataset, user pairs labeled as friend, family, or romantic partner were considered positive examples of close social relationships. In the CNS dataset, friendship relationships declared through the Facebook platform were used as positive labels. Table 1 summarizes the main statistics of both datasets.

Table 1. Datasets' statistics.

Dataset	Individuals	Pairs	Positive Pairs	Negative Pairs
CNS	608	880	714	166
NH	10,436	10,859	1933	8926

Table 1 summarizes the datasets' statistics, including the number of distinct participants, total pairs, and the distribution of positive (Label 1) and negative (Label 0) pairs. Communication features were computed for global and directional interactions, also accounting for weekday and weekend variations to capture temporal behavior differences.

4. Methodology

This section describes the proposed methodology for predicting close social relationships between pairs of users based on mobile communication records. The methodological flow, ranging from data collection to the evaluation of research questions, is illustrated in Figure 1.

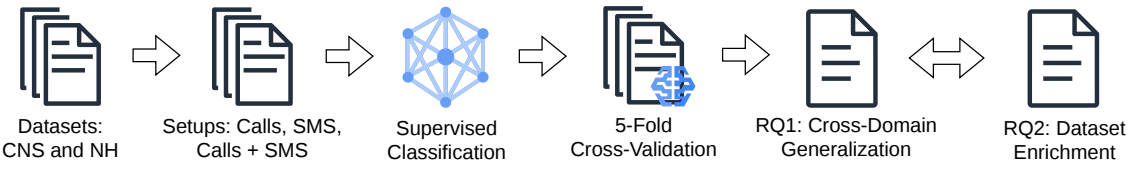


Figure 1. Representation of the research methodological flow.

4.1. Problem Formulation

The task is formulated as a binary classification problem, where the goal is to determine whether a pair of individuals $(p_i, \dot{p}_i)$ exhibits a close social relationship. To conduct the experiments, we define a set of communication setups $C = \{\text{Calls, SMS, Calls + SMS}\}$ and a set of databases $D = \{\text{NetHealth (NH), Copenhagen Networks Study (CNS)}\}$.

Formally, for each pair $(c, d) \in C \times D$, let $\mathcal{S}_d^c = \{(p_i, \dot{p}_i, \mathbf{x}_i^c, y_i)\}_{i=1}^N$ denote the corresponding subset of data. In this formulation:

- $(p_i, \dot{p}_i)$ represent the two interlocutors in the pair;

- $y_i \in \{0, 1\}$ is the class label, where $y_i = 1$ denotes a declared positive relationship and $y_i = 0$ represents its absence;
- $\mathbf{x}_i^c = \phi(p_i, \dot{p}_i, c)$ is the feature vector extracted from their communication history, which varies strictly according to the selected communication setup c.

The variation of $\mathbf{x}_i^c$ occurs as follows: if $c =$ Calls, the vector contains only phone call attributes; if $c =$ SMS, it contains only message attributes; and if $c =$ Calls + SMS, attributes from both domains are combined into a single feature space. The model f learns a mapping $f(\mathbf{x}_i^c) \rightarrow \hat{y}_i$ to predict the pair's class.

Three training strategies are evaluated for each communication setup $c \in C$: M_{NH}^c, trained exclusively on $\mathcal{S}_{\text{NH}}^c$; M_{CNS}^c, trained exclusively on $\mathcal{S}_{\text{CNS}}^c$; and M_{Super}^c, trained on the concatenation $\mathcal{S}_{\text{NH}}^c \cup \mathcal{S}_{\text{CNS}}^c$. This organization enables the analysis of cross-domain generalization (RQ1) and dataset enrichment (RQ2).

4.2. Feature Engineering

For each pair $(p_i, \dot{p}_i)$, we construct a feature vector $\mathbf{x}_i^c$ composed of communication statistics computed globally, directionally ($p_i \rightarrow \dot{p}_i$ and $\dot{p}_i \rightarrow p_i$), and segmented between weekdays and weekends to capture behavioral nuances.

Temporal Encoding. Since time-of-day variables are cyclic, temporal attributes are transformed using trigonometric encoding. A full day is organized into $86,400$ seconds. For any temporal variable t within this cycle, we compute:

$$t_{\sin} = \sin\left(2\pi \frac{t}{86400}\right) \quad \text{and} \quad t_{\cos} = \cos\left(2\pi \frac{t}{86400}\right)$$

This preserves the cyclic nature of time, capturing similarities between nearby times such as 23:59 and 00:01. After encoding, the original values are removed.

Call and SMS Features. The feature vector $\mathbf{x}_i^c = \phi(p_i, \dot{p}_i, c)$ is constructed according to the selected communication setup c. When $c \in \{\text{Calls}, \text{Calls + SMS}\}$, the extracted attributes comprise the total number of calls, average, minimum, and maximum call duration, the timestamps of the earliest and latest interactions, and the number of distinct active days. Similarly, when $c \in \{\text{SMS}, \text{Calls + SMS}\}$, analogous statistics are extracted, including total message volume, the timestamps of the first and last messages, and the number of distinct active days. In both cases, features are computed globally, directionally, and partitioned according to temporal context.

5. Experimental Evaluation

This section presents the experimental methodology adopted to evaluate close social relationship classification and the impact of different communication modalities and training strategies on cross-domain generalization and dataset enrichment.

5.1. Experimental Setup

The prediction task is formulated as a binary classification problem in which each pair of users receives the label **1** when a close social relationship is reported and **0** otherwise. The classifier adopted in this work is LightGBM (*Light Gradient Boosting Machine*)

Table 2. Metrics by scenario and training location.

Metrics	CALLS			SMS			CALLS + SMS		
	M_{CNS}	M_{NH}	M_{Super}	M_{CNS}	M_{NH}	M_{Super}	M_{CNS}	M_{NH}	M_{Super}
NetHealth dataset (NH)									
Accuracy	0.227 ± 0.029	$\mathbf{0.858 \pm 0.008}$	0.811 ± 0.008	0.223 ± 0.023	$\mathbf{0.855 \pm 0.007}$	0.807 ± 0.006	0.216 ± 0.033	$\mathbf{0.870 \pm 0.005}$	0.821 ± 0.003
F1-Mic	0.227 ± 0.029	$\mathbf{0.858 \pm 0.008}$	0.811 ± 0.008	0.223 ± 0.023	$\mathbf{0.855 \pm 0.007}$	0.807 ± 0.006	0.216 ± 0.033	$\mathbf{0.870 \pm 0.005}$	0.821 ± 0.003
F1-Mac	0.210 ± 0.024	$\mathbf{0.716 \pm 0.013}$	0.661 ± 0.012	0.208 ± 0.017	$\mathbf{0.694 \pm 0.013}$	0.641 ± 0.011	0.199 ± 0.028	$\mathbf{0.751 \pm 0.008}$	0.690 ± 0.005
F1-Wgh	0.143 ± 0.027	$\mathbf{0.846 \pm 0.011}$	0.785 ± 0.011	0.145 ± 0.016	$\mathbf{0.837 \pm 0.008}$	0.775 ± 0.006	0.129 ± 0.028	$\mathbf{0.862 \pm 0.007}$	0.801 ± 0.005
Copenhagen Networks Study dataset (CNS)									
Accuracy	$\mathbf{0.773 \pm 0.026}$	0.247 ± 0.014	0.287 ± 0.015	$\mathbf{0.784 \pm 0.029}$	0.179 ± 0.007	0.225 ± 0.006	$\mathbf{0.770 \pm 0.023}$	0.237 ± 0.017	0.277 ± 0.016
F1-Mic	$\mathbf{0.773 \pm 0.026}$	0.247 ± 0.014	0.287 ± 0.015	$\mathbf{0.784 \pm 0.029}$	0.179 ± 0.007	0.225 ± 0.006	$\mathbf{0.770 \pm 0.023}$	0.237 ± 0.017	0.277 ± 0.016
F1-Mac	$\mathbf{0.493 \pm 0.024}$	0.240 ± 0.018	0.271 ± 0.019	$\mathbf{0.455 \pm 0.020}$	0.153 ± 0.006	0.185 ± 0.004	$\mathbf{0.499 \pm 0.033}$	0.226 ± 0.021	0.257 ± 0.021
F1-Wgh	$\mathbf{0.728 \pm 0.040}$	0.193 ± 0.028	0.213 ± 0.027	$\mathbf{0.719 \pm 0.044}$	0.057 ± 0.005	0.086 ± 0.004	$\mathbf{0.728 \pm 0.043}$	0.170 ± 0.034	0.192 ± 0.032

[Ke et al. 2017], selected due to its efficiency on structured data and robustness to heterogeneous feature spaces. The model was trained using 300 estimators, learning rate of 0.05, 63 leaves per tree, maximum tree depth of 6, and random seed fixed at 42. The optimization objective is binary classification using binary log-loss.

Evaluation follows a stratified 5-fold cross-validation protocol with an 80/20 train-test split in each round, preventing data leakage. For the combined-domain setup ($c =$ Calls + SMS), partitions from both datasets are concatenated within each fold. The reported metrics include accuracy and F1-score. Results are presented as mean $\pm$ standard deviation across the five splits to assess both predictive performance and model stability.

5.2. Results

Table 2 summarizes the results obtained for each communication setup and training strategy. The analysis focuses on the two proposed research questions: (RQ1) whether communication datasets generalize across domains, and (RQ2) whether combining datasets from different communication environments improves predictive performance.

Overall, the results reveal a strong domain dependency in communication behavior. Models trained and evaluated on the same dataset consistently achieved the best performance, while cross-domain evaluation led to substantial degradation across all communication setups. Furthermore, combining datasets during training did not improve the results and frequently reduced the performance obtained by single-domain models.

Another relevant observation is that the multimodal Calls + SMS setup generally achieved the best in-domain performance, particularly for NH. This suggests that integrating complementary communication modalities provides richer behavioral signals for identifying close social relationships when training and testing data follow similar distributions.

RQ1 – Cross-domain generalization. The experimental results indicate that communication datasets exhibit limited cross-domain generalization. In all communication setups, models achieved the best performance when trained and evaluated on the same dataset, while models transferred across datasets experienced substantial performance degradation. This behavior was consistently observed for both NH and CNS, suggesting that communication patterns associated with close social relationships are strongly dependent on the characteristics of each dataset and communication environment.

Additionally, the cross-domain models presented a considerable reduction in F1-

based metrics, indicating difficulties in maintaining balanced predictive performance across classes. These findings suggest that behavioral and interaction patterns learned from one communication dataset do not transfer effectively to another domain, even when the prediction task remains the same.

RQ2 – Effect of combining datasets from different communication environments. The results indicate that combining datasets during training does not improve predictive performance. Across all communication setups, the combined-domain model (M_{Super}) consistently underperformed compared to the best single-domain model trained directly on the target dataset.

Although the combined model occasionally achieved slight improvements over purely cross-domain training, these gains were insufficient to match the performance of native-domain models. This behavior suggests that simply increasing data volume and diversity does not compensate for the domain discrepancies between communication datasets. Overall, the results reinforce the existence of strong domain dependency in communication behavior, limiting the effectiveness of straightforward dataset combination strategies for close social relationship classification.

6. Conclusion

This work investigated close social relationship classification using mobile communication data from two public datasets collected in different environments. Experiments were conducted to evaluate both cross-domain generalization and the effect of combining heterogeneous datasets under three communication setups: Calls, SMS, and Calls + SMS.

The results show that communication dynamics associated with close social relationships are strongly domain-sensitive. Models trained on one communication environment were unable to generalize effectively to another dataset, presenting significant performance degradation in cross-domain evaluations. In addition, combining heterogeneous datasets during training did not improve predictive performance, indicating that increasing data diversity alone is insufficient to overcome domain differences.

Finally, the multimodal Calls + SMS setup generally achieved the best in-domain results, suggesting that combining communication modalities provides complementary information for social relationship prediction.

Limitations and Future Work. One limitation of this study is the small number of publicly available datasets containing both communication records and self-reported social relationships. Moreover, the analyzed datasets differ in population characteristics and labeling strategies, which may intensify domain discrepancies. As future work, we intend to investigate graph-based and temporal modeling techniques to improve cross-domain robustness.

Declaration of Generative AI use. Generative AI was used only for linguistic refinement and grammatical editing. The authors conducted all scientific research and data analysis.

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